Global multi-crop agricultural trial data supported by citizen science

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Abstract

We present a harmonized global dataset generated through the tricot (triadic comparisons of technology options) citizen science approach, comprising decentralized on-farm agricultural experiments conducted between 2011 and 2025 across more than 25 countries and over 30 crop species. The dataset represents the outcome of a long-term effort to develop a FAIR-by-design infrastructure for agricultural experimentation, integrating standardized protocols, digital data collection, controlled vocabularies, environmental covariates, socio-economic information, and links to external repositories. The resulting resource provides a unique foundation for studying farmer preferences, genotype-by-environment interactions, climate adaptation, technology adoption, and participatory breeding at unprecedented scale.

Keywords: digital innovation, FAIR data, tricot approach, participatory evaluation

Introduction

Agricultural innovation increasingly requires decentralized, context-specific evidence generated directly in the target environments [1, 2]. The tricot (triadic comparison of technology options) approach was developed to address this challenge by enabling large-scale citizen science experimentation in agriculture [3, 4]. Early implementations in Ethiopia, India, and Central America demonstrated that farmers could generate robust experimental data through simple, distributed trial designs [3, 5, 1, 6].

Briefly summarize:

- Origins of tricot [3, 4].
- First pilot studies [5, 1, 7]
- Citizen science rationale [8, 9].

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Data base overview

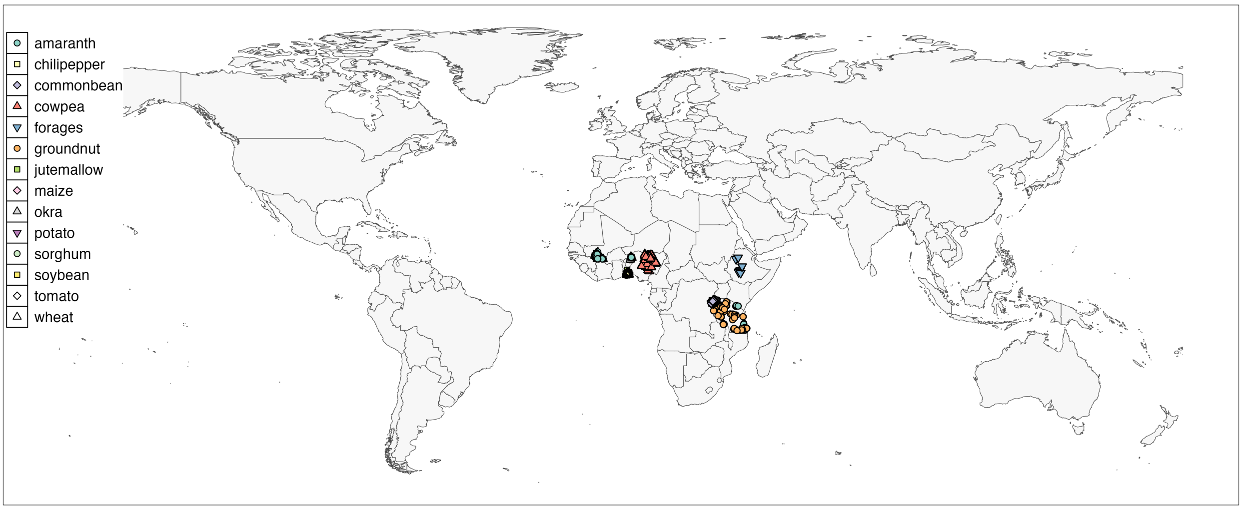

Figure 1: Geographical distribution of crops and trial sites that are currently available for public use. Each point represents a trial location, and the legend indicates the crop(s) associated with those sites. Administrative boundaries are based on the Database of Global Administrative Areas (GADM, https://gadm.org). The authors remain neutral with regard to jurisdictional claims in maps.

Describe the repository structure:

- GitHub repository.
- Zenodo archive.
- Dataset organization.
- Versioning strategy.

Describe the coverage. Table(s) describing the countries, temporal coverage and crops

Crop	Country	Years	Participants

Data set statistics

Indicator	Value
Countries	XX
Crops	XX
Studies	XX
Participants	XX
Observations	XX
Organizations	XX

The tricot research infrastructure

Over the last fifteen years, tricot evolved from a participatory experimental method into a global digital FAIR-by-design infrastructure supporting decentralized agricultural experimentation.

Short description of the development phases, for example Phase I (2011–2016) where we tested tricot as proof of concept with pilot trials in Ethiopia, India, Central America [3], early digital tools [3] and initial climate adaptation applications [5].

Then Phase II (2017–2022) on the digitalization and scaling, with the new version of ClimMob [10], expansion across countries, crops and contexts [11, 12, 7, 13, 14], and integration into breeding programs [15, 16, 17, 18].

Then finally to institutionalization and digital interoperability in Phase III (2022–Present), standardized metadata, ontology mapping, SOP development, and interoperability with external systems [19].

maybe this could be a different section…

To ensure consistency across crops, countries, and implementing organizations, all studies are implemented using standardized protocols.

Describe:

- SOP framework.
- Crop-specific protocols.
- Multilingual documentation.
- Training materials.
- Version control.
- Zenodo publication of protocols.

Include links to published SOPs maybe in a table.

The tricot open data community

The data insfrastuscture represents a collaborative effort involving CGIAR centers, national agricultural research systems, NGOs, universities, and farmer communities. The repository is intended as a living resource that will continue to expand through future contributions.

Methods

Should answer three questions, how data is collected? how it is collated and curated? how it is made available?

Experimental design

Describe the experimental design

- Triadic comparison of technology options.
- Incomplete block design.
- Random allocation of technologies.
- Farmer-led observations.
- Ranking-based evaluations.

Describe types of studies and contexts

- On-farm variety testing.
- Consumer preference testing.
- Product concept testing.
- Seed system studies.
- Participatory breeding.

Data curation

Describe metadata standards and ontology mapping
Describe trait standardization

- Crop Ontology.
- Trait definitions.
- Observation methods.
- Scales and units.

Describe metadata standards

- DataCite.
- ROR identifiers.
- Funding metadata.
- Contributor metadata.

Data architecture

The repository follows a hierarchical structure linking study metadata, participant-level information, plot-level observations, and analytical outputs.

Describe the key elements of the json file metadata, plot data, block data, analysis.

Integration with external data sources

The repository is designed to support linkage with external data systems. Describe potential integration with climate date (CHIRPS, Era5, etc), socio-economic data (RHoMIS, household surveys), genetic resources systems (Genesys), and other data insfrastructures (Fairgrounds, etc).

Case studies

- Variety recommendation.
- Climate resilience studies.
- Decentralized breeding.
- Product profile development.
- Trait prioritization.
- Consumer testing.

Ethics and data governance

- Agricultural data governance.
- FAIR implementation assessments.
- Who owns the data.
- How we address issues of traditional knowledge.
- Why we chose the licence CC0.

Discussion

Lessons learned from building FAIR agricultural citizen science infrastructure
Standardization is both technical and social

- Ontology mapping.
- Partner alignment.
- Governance processes.

Interoperability requires long-term investment

- Metadata harmonization.
- Repository development.
- Cross-institutional coordination.

FAIR is a process rather than an endpoint

- Continuous improvement.
- Dataset expansion.
- Community contributions.

Future development

Code availability

Code is available on GitHub https://github.com/AgrDataSci/tricot-data

Data availability

Data is fully available in Zenodo (https://doi.org/10.5281/zenodo.17112491) under the CC0 License.

Author contributions

Use CRediT taxonomy.

Competing interests

The authors declare no competing interests.

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